Common digital certificate systems are X.509 and PGP.

- Pretty Good Privacy (PGP): Phil Zimmermann developed PGP in 1991. It is a decentralized encryption program that provides cryptographic privacy and authentication for data communication. PGP encryption uses a serial combination of hashing, data compression, symmetric-key cryptography and asymmetric-key cryptography and works on the concept of "web of trust".
- The X.509 system is a centralized system in which the authenticity of the key is guaranteed by the hierarchy of certification authorities formally certifying the key relationship with the identity of its owner. Due to its clear responsibility, it is easier to implant in the law, X.509 is currently world wide accepted certification technology.



The digital certificate are being issued by a licensed Certifying Authority (CA). NIC, Safescript, TCS, MTNL, e-Mudhra are some of the authorized Certifying Authorities under Government of India.

Digital signature

A digital signature is a mechanism that is used to verify that a particular digital document, message or transaction is authentic.

It provides the receiver the guarantee that the message was actually generated by the sender. It also confirms

Digital signature	Digital certificate
A digital signature is a mechanism that is used to verify that a particular digital document, message or transaction is authentic.	A digital certificate is a computer file which officially approves the relation between the holder of the certificate and a particular public key.
Digital signatures are used to verify the trustworthiness of the data being sent	Digital certificates are used to verify the trustworthiness of the sender.
Digital signature is to ensure that a data remain secure from the point it was issuedand it was not modified by a third party.	Digital certificate binds a digital signature to an entity
It provides authentication, non-repudiation and integrity	It provides authentication and security.
A digital signature is created using a Digital Signature Standard (DSS). It uses a SHA-1 or SHA-2 algorithm for encrypting and decrypting the message.	A digital certificate works on the principles of public key cryptography standards (PKCS). It creates certificate in the X.509 or PGP format.
The document is encrypted at the sending end and decrypted at the receiving end using asymmetric keys.	A digital certificate consist of certificate's owner name and public key, expiration date, a Certificate Authority 's name , a Certificate Authority's digital signature

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